LEARNER GUIDE

Inequality symbols and double-ended inequalities

Inequalities · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Most questions so far have had **one** answer. But real life is full of **boundaries**: you must be *at least* 12 to ride; a bag must weigh *less than* 7 kg. A boundary does not pick out one number — it lets a whole **set** of numbers through. We write boundaries with four symbols: < (is less than), > (is greater than), ≤ (is less than or equal to) and ≥ (is greater than or equal to). A **double-ended** inequality such as $-1 < x \leq 3$ traps a number between two boundaries.

What you are learning to notice

- On a number line, numbers get **greater** as you move to the **right** — so –2 is greater than –5.
- a < b says the same as b > a. The open side of the symbol faces the greater number.
- At a boundary, the symbol decides: ≤ and ≥ **include** the boundary; < and > **leave it out**.
- An inequality describes a whole **set** of numbers, which you can test one number at a time.

Moving through it, one step at a time

To decide whether a number obeys an inequality, put it into the rule and ask: is the statement true? For a double-ended inequality, check both parts. Take extra care at the boundary numbers: look at the symbol next to the boundary — ≤ or ≥ lets it in, < or > keeps it out. To list the **whole numbers** (integers) that obey a rule, start at the lower boundary, decide whether it is in, step up one at a time (remembering 0), and decide about the upper boundary.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Through the gate?

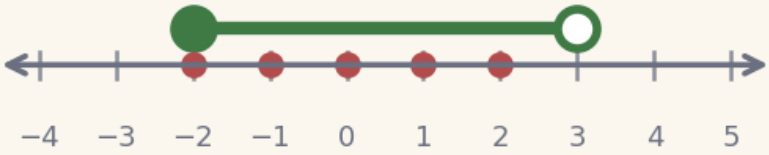
The gate's rule is $x \geq -2$. Do –2, –3 and 0 get through?

1. Test –2: is $-2 \geq -2$? It is equal, and ≥ includes equal.	–2 gets through
2. Test –3: is $-3 \geq -2$? –3 is to the left of –2, so it is smaller.	–3 does not
3. Test 0: is $0 \geq -2$? 0 is to the right of –2.	0 gets through

–2 and 0 get through; –3 does not. The filled circle at –2 shows that the boundary is included.

Example 2 — Listing whole numbers

List the whole numbers n with $-2 \leq n < 3$.



1. Lower boundary: ≤ includes –2.	start at –2
2. Step up one at a time, remembering 0.	–2, –1, 0, 1, 2
3. Upper boundary: < leaves 3 out.	stop before 3

–2, –1, 0, 1, 2 — five whole numbers. Filled circle at –2 (included), open circle at 3 (left out).

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (ordering numbers, including negative numbers, on a number line), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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